CAPI overview

1. Custom defined accelerator within processor context
2. Function based accelerator
   (a) Main application executed on host processor
   (b) Computed heavy function offloaded
   (c) Single binary with both hardware and software versions of function
      ➤ Runs without accelerator
3. CAPI device: Full peer to processor
   (a) Enables memory between accelerator function unit (AFU) and application
      ➤ Via power service layer, AFU can understand application's virtual address
      ➤ No need to pin application memory for I/O
      ➤ AFU works within application context

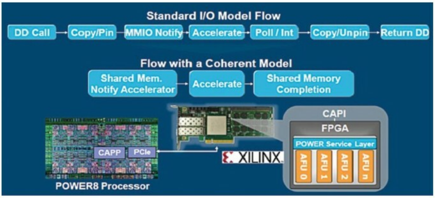

Fig. 8: Coherent accelerator processor interface

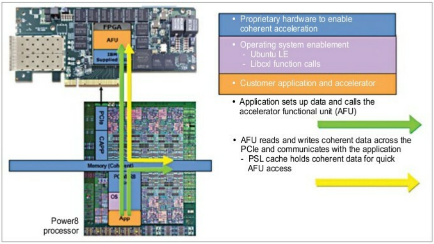

Fig. 9: CAPI technology connections

basic designs: a 24 SMT4 processor and a 12 SMT8 processor. The 24 SMT4 processor will be optimised for the Linux ecosystem and will target web service companies such as Google, which need to run across several thousand machines. It will feature four threads.

The 12 SMT8 processor will be optimised for the Power ecosystem and will target larger systems designed for running big data or artificial intelligence (AI) applications. It will feature eight threads.

Both designs will come in two models: The scale-out model will come with two CPU sockets on the motherboard, while the scale-up model will come with multiple CPU sockets.

Power9 processor will be a 14nm high-performance FinFET product. It will directly attach to DDR4 RAM, connect to peripherals and Nvidia GPUs via PCIe Gen 4 and NVLink 2.0, and transfer accelerator data at 25Gbps speeds. Power9 processor will have multiple connectors to attach FPGAs, GPUs and ASICs. It will be well suited for AI, cognitive computing, analytics, visual computing and hyper-scale web serving. IBM will exclusively use Power9 in its servers and make it available to other hardware companies by licensing the design to them. Power9 processors can be attached directly to various accelerators rather than communicating via the PCIe bus.

Heterogeneous computing approaches for cloud data centres

In heterogeneous systems, different kinds of processors—x86 CPUs and FPGAs—cooperate on a computing task. Heterogeneous architectures are already widely used in the mobile world, where ARM CPUs, GPUs, encryption engines and digital signal processing (DSP) cores routinely work together, often on a single die.

There are two approaches to establish heterogeneous computing systems in cloud data centres.

The first approach is based on heterogeneous super nodes that tightly couple compute resources to multi-core CPUs and their coherent memory system via high-bandwidth, low-latency interconnects like CAPI or NVLink. CAPI is encapsulated by the PCIe and appears to use the same bus architecture with higher bandwidth capabilities and lower overall latencies. NVLink is a high-bandwidth, energy-efficient interconnect that enables ultra-fast communication between the CPU and GPU, and between GPUs. The technology allows data sharing at speeds five to twelve times faster than the traditional PCIe Gen 3 interconnect, resulting in dramatic speed-ups in application performance and creating a new